

MAT1002 Lecture 13,

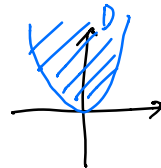
Outline

- Functions of several variables
- Limits and continuity
- Partial derivatives of $f(x,y)$

Functions of Several Variables

e.g. $f(x,y) = kx^{1/3}y^{2/3}$, $D = \mathbb{R}^2$.

$f(x,y) = \sqrt{y-x^2}$, $D = \{(x,y) : y-x^2 \geq 0\} = \{(x,y) : y \geq x^2\}$.



- In general, $f(x_1, \dots, x_n)$ function with
 n real variables, $D \subseteq \mathbb{R}^n$.

- Alternative notation: $f(\vec{x}) = f(x_1, \dots, x_n)$
 $\vec{x} \in \mathbb{R}^n$
 vector notation is used to emphasize that there are multiple variables.

- We focus on real-valued functions (scalar function) with two or three variables, which are $f: D \rightarrow \mathbb{R}$, $D \subseteq \mathbb{R}^2$ or $D \subseteq \mathbb{R}^3$.

- Visualization: $\mathbb{R}^2$ $\mathbb{R}$

Definition

Let D be a subset of $\mathbb{R}^n$.

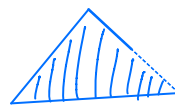
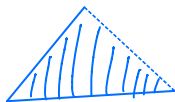
- A point P in $\mathbb{R}^n$ is called an **interior point** of D if some open ball centered at P completely lies in D .
- A point P in $\mathbb{R}^n$ is called a **boundary point** of D if every open ball centered at P intersects both D and $\mathbb{R}^n \setminus D$.
- The set D is said to be **open** if every point in D is an interior point of D ; it is said to be **closed** if it contains all of its boundary points.
- The set D is said to be **bounded** if it lies in some ball with finite radius; it is said to be **unbounded** otherwise.

$\leftarrow$ in $\mathbb{R}^2$, "open disk"

disk in $\mathbb{R}^2$

Remark Open, closed and bounded sets in $\mathbb{R}^n$ are analogous to open, closed and bdd intervals in $\mathbb{R}$, but they are more complicated in general.

$[a, b)$

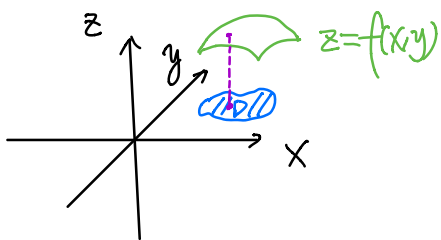


Graphs and Level Curves/Surfaces

Def: The **graph** of $f: D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^n$, is the set
 $\{(x_1, \dots, x_n, f(x_1, \dots, x_n)) : (x_1, \dots, x_n) \in D\}$.

Hence, the graph of an n -variable function lie in $\mathbb{R}^{n+1}$.

e.g. $n=2$; $z = f(x, y)$



Think of climbing a mountain:

(x, y) : 2D position on ground level

$f(x, y)$: mountain altitude at (x, y)

e.g. $n=3$; $w = f(x, y, z)$, graph is

$\{(x, y, z, w) : w = f(x, y, z), (x, y, z) \in D\}$.

Can think of temperature: (x, y, z) : 3D position in space

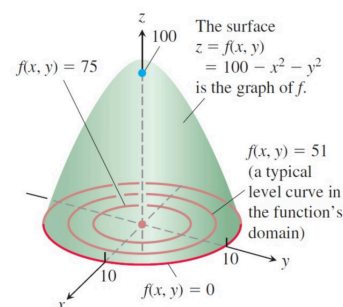
$f(x, y, z)$: temperature at (x, y, z) .

Def: Given $f: D \rightarrow \mathbb{R}$, $D \subseteq \mathbb{R}^n$, and any constant $K \in \mathbb{R}$.

- For $n=2$, the set $\{(x,y) \in D : f(x,y) = K\}$ is called a **level curve** of f .
- For $n \geq 3$, the set $\{(x_1, \dots, x_n) \in D : f(x_1, \dots, x_n) = K\}$ is called a **level surface** of f .

e.g. Describe the level curves of $f(x,y) = 100 - x^2 - y^2$.

Ex. Use GeoGebra to investigate the level curves for various functions.



e.g. Consider the level surfaces of $f(x,y,z) = \sqrt{x^2 + y^2 + z^2}$ and $g(x,y,z) = y^2 - x^2 - z$.

Plot some of them using GeoGebra.

Limits

For a two-variable function f with domain D , we say

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$$

if, $\forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall (x,y) \in D$,

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta \implies |f(x,y) - L| < \epsilon.$$

In general :

Definition

Let $f : D \rightarrow \mathbb{R}$ be a function, where $D \subseteq \mathbb{R}^n$. We say that L is the **limit** of f as $\vec{x}$ approaches $\vec{x}_0$, and write

$$\begin{array}{l} \vec{x} \in \mathbb{R}^n : \\ n \text{ real variables} \end{array} \quad \lim_{\vec{x} \rightarrow \vec{x}_0} f(\vec{x}) = L,$$

if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $\vec{x} \in D$,

$$0 < |\vec{x} - \vec{x}_0| < \delta \text{ implies } |f(\vec{x}) - L| < \epsilon.$$

Remark

Note that $0 < |\vec{x} - \vec{x}_0| < \delta$ is equivalent to $\vec{x} \in B_\delta(\vec{x}_0) \setminus \{\vec{x}_0\}$.

open ball centered at $\vec{x}_0$ with radius δ .

The Most Basic Limits

Using the definition of limits, it can be shown that

$$\lim_{(x,y) \rightarrow (x_0,y_0)} x = x_0, \quad \lim_{(x,y) \rightarrow (x_0,y_0)} y = y_0, \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} k = k,$$

where k is a constant.

- Similarly, $\lim_{(x,y,z) \rightarrow (x_0,y_0,z_0)} z = z_0$, etc.
- Note that $\lim_{(x,y) \rightarrow (x_0,y_0)} x$ means $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y)$, where $f(x,y) = x$. $\leftarrow$ Graph is $\{(x,y,z) : z = x\}$.

Proof of $\lim_{(x,y) \rightarrow (x_0,y_0)} x = x_0$:

- Let $\epsilon > 0$. Want to show that $\exists \delta > 0$ s.t.

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta \Rightarrow |x-x_0| < \epsilon. \quad (*)$$

• Choose $\delta := \varepsilon$. Suppose $0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$. Then

$$|x-x_0| = \sqrt{(x-x_0)^2} \leq \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta = \varepsilon.$$

• (*) is proven. □

Properties of Limits

The following properties, although only stated for two-variable functions, hold for functions with finitely many variables.

THEOREM — Properties of Limits of Functions of Two Variables

The fol-

lowing rules hold if L , M , and k are real numbers and

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M.$$

1. Sum Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) + g(x,y)) = L + M$$

2. Difference Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) - g(x,y)) = L - M$$

3. Constant Multiple Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} kf(x,y) = kL \quad (\text{any number } k)$$

4. Product Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) \cdot g(x,y)) = L \cdot M$$

5. Quotient Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}, \quad M \neq 0$$

6. Power Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)]^n = L^n, \quad n \text{ a positive integer}$$

7. Root Rule:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n}$$

if n even, assume $f(x,y) \geq 0$
for all (x,y) in some $B_\delta(x_0,y_0)$.

e.g. $\lim_{(x,y) \rightarrow (4,8)} 5x^2y^{1/3} = \left(\lim_{(x,y) \rightarrow (4,8)} 5 \right) \left(\lim_{(x,y) \rightarrow (4,8)} x^2 \right) \left(\lim_{(x,y) \rightarrow (4,8)} y^{1/3} \right)$

$$= 5 \left(\lim_{(x,y) \rightarrow (4,8)} x \right)^2 \left(\lim_{(x,y) \rightarrow (4,8)} y \right)^{1/3} = 5 \cdot 4^2 \cdot 8^{1/3} = 160.$$

e.g. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}} = 0$. Note that domain of function is

$$D = \{(x,y) : x \geq 0, y \geq 0, x \neq y\}.$$

Limits Along Paths

For a one-variable function, we know that $\lim_{x \rightarrow x_0} f(x) = L$ if and only if

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = L.$$

An analogous statement for functions with multiple variables is the following.

Theorem

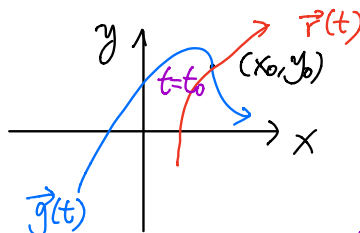
Let $f : D \rightarrow \mathbb{R}$ be a function, where $D \subseteq \mathbb{R}^n$. Then $\lim_{\vec{x} \rightarrow \vec{x}_0} f(x) = L$ if and only if

f converges to L as $\vec{x}$ approaches $\vec{x}_0$ along any path in D .

Curve

For function with two variable:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y)$$



Formally, if $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$ and $\vec{r}(t)$ is any curve in $\mathbb{R}^2$ with $\lim_{t \rightarrow t_0} \vec{r}(t) = \langle x_0, y_0 \rangle$, then $\lim_{t \rightarrow t_0} f(\vec{r}(t)) = L$.

A consequence of the theorem is the following: if f does not converge to the same number along two different paths, then $\lim_{\vec{x} \rightarrow \vec{x}_0} f(x)$ does not exist.

Curves

e.g. For $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$:

- On x -axis ($y=0, x \neq 0$), $f(x,y) \equiv 1$.
- On y -axis ($x=0, y \neq 0$), $f(x,y) \equiv -1$.
- Hence $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ D.N.E.

Example

Show that the following limits do not exist.

(a) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$.

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$.

Sol: (a) Along the curve $y=kx$, $(x,y) \rightarrow (0,0)$ as $x \rightarrow 0$.

$$\text{So } \lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=kx}} \frac{xy}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x(kx)}{x^2 + (kx)^2} = \lim_{x \rightarrow 0} \frac{k \cancel{x^2} \text{ Same}}{(1+k^2)\cancel{x^2}} = \frac{k}{1+k^2}$$

Along $y=x$, limit $= \frac{1}{2}$; along $y=-x$, limit $= -\frac{1}{2}$.

So limit D.N.E.

(b) Along the curve $x=ky^2$, $(x,y) \rightarrow 0$ as $y \rightarrow 0$, so

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } x=ky^2}} \frac{xy^2}{x^2 + y^4} = \lim_{y \rightarrow 0} \frac{k \cancel{y^4} \text{ Same}}{(k^2+1)\cancel{y^4}} = \frac{k}{k^2+1} = \begin{cases} \frac{1}{2}, & \text{if } k=1 \\ -\frac{1}{2}, & \text{if } k=-1. \end{cases}$$

Hence limit D.N.E.

Squeeze (Sandwich) Theorem

Theorem: If $g(x,y) \leq f(x,y) \leq h(x,y)$, $\forall (x,y) \in B_r(a,b)$, $r > 0$, and

$$\lim_{(x,y) \rightarrow (a,b)} g(x,y) = \lim_{(x,y) \rightarrow (a,b)} h(x,y) = L,$$

then

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L.$$

Proof: Exercise.

e.g. Find $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2+y^2}$. (Do this in at least two ways.)

Continuity

Definition

Let $f : D \rightarrow \mathbb{R}$ be a function, where $D \subseteq \mathbb{R}^n$, and let $\vec{x}_0 \in D$. Then f is said to be **continuous at $\vec{x}_0$** if

$$\lim_{\vec{x} \rightarrow \vec{x}_0} f(\vec{x}) = f(\vec{x}_0).$$

We say that f is **continuous** if f is continuous at $\vec{x}_0$ for every $\vec{x}_0 \in D$.

A consequence of properties of limits is that sums, differences, products, quotients and powers of continuous functions are continuous everywhere they are defined.

Hence polynomials, such as
 $f(x,y,z) = 48z^3 - 61xyz + 13z - 7$, are cts.

Example

The function defined by

$$f(x, y) := \begin{cases} \frac{3x^2y}{x^2+y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

is continuous on $\mathbb{R}^2$.

Compositions of Continuous Functions

Just like functions with one variable, the composition of two continuous functions is continuous. In particular:

If $f : D \rightarrow \mathbb{R}$ is a multiple-variable continuous function, and g is a single-variable continuous function whose domain contains the range of f , then $g \circ f$ is a continuous function on D .

e.g. $f(x, y) = \cos \frac{xy}{x^2+1}$ is continuous on $\mathbb{R}^2$, and

$g(x, y, z) = e^{xyz} \cos z$ is cts on $\mathbb{R}^3$.

Partial Derivatives for Two-Variable Functions

Definition

Let $f : D \rightarrow \mathbb{R}$ be a function, where $D \subseteq \mathbb{R}^2$, and let (x_0, y_0) be an interior point of D . The **partial derivative** of f at (x_0, y_0) **with respect to x** , denoted by $\frac{\partial}{\partial x} f(x_0, y_0)$, is defined by

Also:

$$\frac{\partial f}{\partial x}(x_0, y_0)$$

$$\left(\frac{\partial}{\partial x} f(x_0, y_0) \right) := \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

and the **partial derivative** of f at (x_0, y_0) **with respect to y** , denoted by $\frac{\partial}{\partial y} f(x_0, y_0)$, is defined by

$$\frac{\partial}{\partial y} f(x_0, y_0) := \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}.$$

$\frac{d}{dx} g(x)$, where
 $g(x) := g_{y_0}(x)$
 $:= f(x, y_0)$

Alternative notations for $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ include:

$$\frac{\partial f}{\partial x}, f_x, D_x f \text{ and } D_1 f; \quad \frac{\partial f}{\partial y}, f_y, D_y f \text{ and } D_2 f.$$

